NMOS Transistor Equations:

Saturation (or active) region: $i_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(v_{GS} - V_{tn})^2(1 + \lambda v_{DS})$

Triode region: $i_D = \mu_n C_{ox}\frac{W}{L}\left[(v_{GS} - V_{tn})v_{DS} - \frac{v_{DS}^2}{2}\right]$

Deep triode region: $r_{DS} = \frac{1}{\mu_n C_{ox}\frac{W}{L}(v_{GS} - V_{tn})}$

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Results from the uploaded $I_D - V_D$ data

The positive-V_D curves in `ID_VD_Curves.csv` are perfectly linear. For example, at $V_G = 3\ \mathrm{V}$:

V_D	I_D
$1\ \mathrm{V}$	$90.8\ \mu\mathrm{A}$
$2\ \mathrm{V}$	$181.6\ \mu\mathrm{A}$
$3\ \mathrm{V}$	$272.4\ \mu\mathrm{A}$
$4\ \mathrm{V}$	$363.2\ \mu\mathrm{A}$
$5\ \mathrm{V}$	$454.0\ \mu\mathrm{A}$

Therefore,

$$\frac{I_D}{V_D} = 90.8\ \mu\mathrm{S}$$

at $V_G = 3\ \mathrm{V}$. This indicates that the present `tft_n1` model behaves as a gate-controlled resistor:

$$I_D = V_{DS} g_{ch}(V_{GS})$$

rather than as a MOSFET with triode and saturation regions.

1. Effective $\mu_n C_{ox}$

In deep triode,

$$g_{ch} = \frac{I_D}{V_{DS}} = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{tn})$$

From the CSV, the high-V_G conductances are:

V_G	$g_{ch} = I_D/V_D$
1.3636 V	29.518 μS
1.9091 V	50.300 μS
2.4545 V	72.264 μS
3.0000 V	90.800 μS

Linear fitting gives:

$$g_{ch} \approx 3.773 \times 10^{-5}(V_{GS} - 0.5725)$$

Therefore,

$$\boxed{V_{tn,\mathrm{eff}} \approx 0.573\ \mathrm{V}}$$

and

$$\boxed{\mu_n C_{ox} \frac{W}{L} \approx 3.77 \times 10^{-5}\ \mathrm{A/V^2}}$$

Your netlist does not pass w and l to `tft_n1`. Using its default $W = L = 2\,\mu\mathrm{m}$, so that $W/L = 1$,

$$\boxed{\mu_n C_{ox} \approx 3.77 \times 10^{-5}\ \mathrm{A/V^2}}$$

The fit has $R^2 \approx 0.9989$.

Can μ_n and C_{ox} be found separately?

No. DC $I_D - V_D$ and $I_D - V_G$ curves determine only their product:

$$\mu_n C_{ox}$$

They cannot uniquely determine μ_n and C_{ox} individually.

For $W/L = 1$,

$$\boxed{\mu_n = \frac{3.77 \times 10^{-5}}{C_{ox}}}$$

or

$$\boxed{C_{ox} = \frac{3.77 \times 10^{-5}}{\mu_n}}$$

with SI units:

- μ_n: $\mathrm{m^2/V}$\cdotp$\mathrm{s}$
- C_{ox}: $\mathrm{F/m^2}$

To calculate C_{ox}, oxide thickness and dielectric constant are required:

$$C_{ox} = \frac{\varepsilon_0 \varepsilon_r}{t_{ox}}$$

Then mobility can be calculated from the extracted product.

2. Channel-length modulation parameter λ

For a proper saturation curve,

$$I_D = I_{D0}(1 + \lambda V_{DS})$$

At fixed V_{GS}, fitting the saturation-region data to

$$I_D = mV_{DS} + b$$

would give

$$\lambda = \frac{m}{b}$$

For $V_G = 3\ \mathrm{V}$, the extracted threshold predicts:

$$V_{DS,\mathrm{sat}} = V_{GS} - V_{tn} = 3 - 0.573 \approx 2.43\ \mathrm{V}$$

Thus the region from approximately 2.5 to $5\ \mathrm{V}$ should be saturation. However, the CSV gives:

$$I_D = 90.8 \times 10^{-6} V_{DS}$$

with the fitted intercept approximately zero:

$$b \approx -2.1 \times 10^{-18}\ \mathrm{A}$$

Consequently,

$$\lambda = \frac{m}{b}$$

is mathematically undefined and physically meaningless.

$$\boxed{\lambda \text{ cannot be extracted from the present model/data}}$$

This is not the same as proving $\lambda = 0$. A device with $\lambda = 0$ would show a flat saturation current, while your curves remain proportional to V_{DS}.

Final extracted parameter summary

Parameter	Result
Effective threshold V_{tn}	$\boxed{0.573\ \mathrm{V}}$
$\mu_n C_{ox}(W/L)$	$\boxed{3.77 \times 10^{-5}\ \mathrm{A/V^2}}$
$\mu_n C_{ox}$, assuming $W/L = 1$	$\boxed{3.77 \times 10^{-5}\ \mathrm{A/V^2}}$
μ_n individually	Cannot be determined without C_{ox}
C_{ox} individually	Cannot be determined without oxide information or C–V data

Parameter	Result
λ	Cannot be determined because the model has no saturation region

To extract a valid λ, the Verilog-A model must implement separate triode and saturation equations rather than only $I_D = V_{DS} g_{ch}(V_{GS})$.

Sources